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Late Cretaceous Paleomagnetism of the Tucson Mountains: Implications for Vertical Axis Rotations in South Central Arizona

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The Tucson Mountains of southern Arizona are the site of an Upper Cretaceous caldera from which the rhyolitic Cat Mountain Tuff was erupted at about 72 Ma. Two magnetic units within the Cat Mountain Tuff are distinguished by paleomagnetic data in both the northern and southern Tucson Mountains. The resurgent Amole pluton (≈72 Ma) in the northern Tucson Mountains was emplaced soon after eruption of the Cat Mountain Tuff but cooled and was magnetized after northeastward tilting (50°–85°) of the adjacent caldera-fill sequence. Petrologic and paleomagnetic data indicate that the lower magnetic unit of the Cat Mountain Tuff caps the Silver Bell Mountains to the northwest. A previous paleomagnetic investigation (N = 34) indicates that the Silver Bell Mountains have been rotated clockwise 30° ± 16° (95% confidence level) about a vertical axis relative to cratonic North America. A similar paleomagnetic study of Upper Cretaceous volcanic, volcaniclastic, and intrusive units in the Tucson Mountains (N = 26) indicates that these rocks have been rotated 7° ± 14° clockwise relative to stable North America. A direct comparison of paleomagnetic directions for the lower unit of the Cat Mountain Tuff shows a 17° ± 10° clockwise rotation between the Silver Bell Mountains and the Tucson Mountains which supports the relative accuracy of the absolute rotations determined for these two mountain ranges. Preliminary paleomagnetic directions for middle Tertiary units from the Silver Bell and Tucson Mountains are consistent with clockwise rotation having occurred prior to deposition of these rocks. Clockwise rotation of crustal blocks in southern Arizona likely was associated with strike-slip movement on major northwest trending faults in the region, and this movement may have been related to oblique subduction of oceanic plates along the western continental margin in Late Cretaceous and early Tertiary time. The available paleomagnetic data indicate that rocks in southern Arizona have not remained unrotated with respect to North America since Late Cretaceous time and that vertical axis rotations may have played an important role in the region during Laramide deformation.

INTRODUCTION

This study of the Tucson Mountains supplements a previous study of the Silver Bell Mountains [*Hagstrum and Sawyer*, 1989] to the northwest (Figure 1) and is part of an ongoing paleomagnetic investigation of vertical axis rotations associated with Late Cretaceous to early Tertiary (Laramide) deformation in the Basin and Range province of southern Arizona. Other paleomagnetic studies in the region have concentrated on refining the Mesozoic and early Cenozoic North American apparent polar wander path (APWP) [*Vugteveen et al.*, 1981; *Kluth et al.*, 1982; *May and Butler*, 1986; *May et al.*, 1986] but are based on the assumption that rocks within the Basin and Range province of southern Arizona have remained unrotated with respect to cratonic North America since Jurassic time. In addition to the general consistency of these paleomagnetic data with the North American APWP, the assumption that Mesozoic rocks in southern Arizona have not undergone vertical axis rotations is also thought to be supported by the apparent internal consistency of fracture and dike trends within Laramide and Tertiary rocks in the region [*Rehrig and Heidrick*, 1976; *Heidrick and Titley*, 1982], by the overall similarity of middle Tertiary core complex transport directions [e.g., *Rehrig and Reynolds*, 1980], and by the lack of vertical axis rotations associated with late Cenozoic high-angle Basin and Range-type faulting [*Coney*, 1978]. More recent paleomagnetic investigations of Late Cretaceous and early Tertiary rocks in southern Arizona, however, have shown significant clockwise vertical axis rotations in the Silver Bell Mountains [*Hagstrum and Sawyer*, 1989] and in the Canelo Hills region [*Hagstrum and Lipman*, 1991] with respect to stable North America.

Clockwise rotation (≈30°) of the Silver Bell Mountains is inferred by *Hagstrum and Sawyer* [1989] to have been related to regional Late Cretaceous and early Tertiary northwest trending strike-slip faulting. The Silver Bell Mountains are bounded to the north and south by subvertical faults, and the N75°–80°W Ragged Top fault to the north shows evidence of major strike-slip displacement between ≈70 Ma and ≈26 Ma, although significant dip-slip displacement is also required. In addition, preliminary paleomagnetic data for middle Tertiary rocks in the vicinity of the Silver Bell Mountains imply that clockwise rotation occurred prior to ≈26 Ma [*Hagstrum and Sawyer*, 1989]. Northwest trending subvertical faults with probable strike-slip displacement are a major element of late Mesozoic tectonics in southeastern Arizona [*Titley*, 1976; *Drewes*, 1981], but movement along them is difficult to interpret because it has been recurrent and of diverse sense. *Heidrick and Titley* [1982] suggested that these faults would have been subjected to left-lateral shear during Laramide deformation based on a tectonic stress field for southern Arizona (σ_1 = N75°E) inferred from regional dike and fracture patterns. *Keith and Wilt* [1986] restrict left-lateral movement on these faults to a medial phase (65–55 Ma) of the Laramide orogeny that is also characterized by epizonal plutonism and porphyry copper mineralization. Recently, Mesozoic porphyry copper deposits across southern Arizona have been interpreted to be widely associated with eruptive centers of arc-related silicic volcanism [*Lipman and Sawyer*, 1985].

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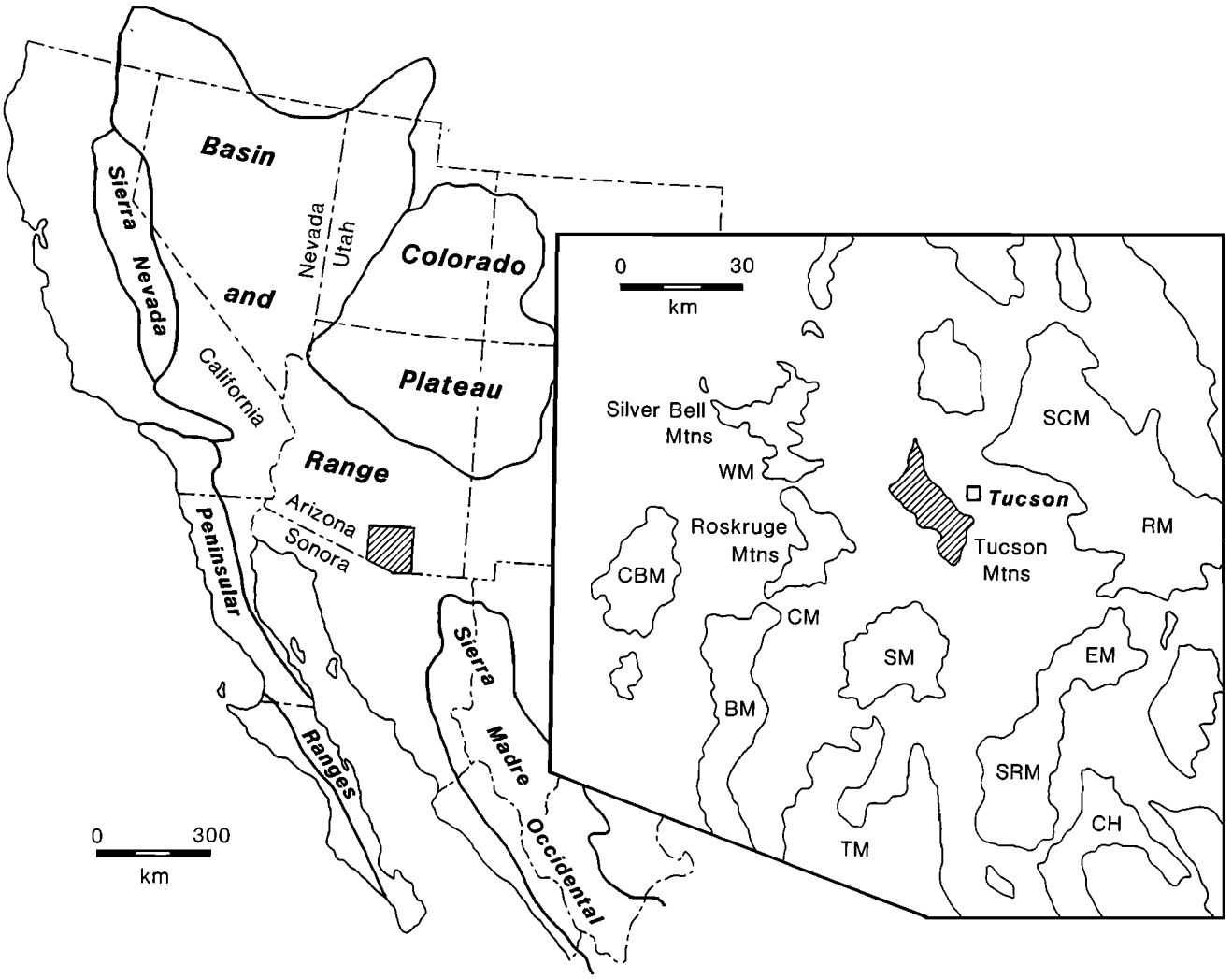


Fig. 1. Regional map (left) showing physiographic provinces in southwestern North America [after *Stewart*, 1978], and index map (right) showing location of the Tucson Mountains (32.2°N, 248.9°E) in south central Arizona. Mountain range abbreviations are BM, Baboquivari Mountains; CBM, Comobabi Mountains; CH, Canelo Hills; CM, Coyote Mountains; EM, Empire Mountains; RM, Rincon Mountains; SCM, Santa Catalina Mountains; SM, Sierrita Mountains; SRM, Santa Rita Mountains; TM, Tumacacari Mountains; and WM, Waterman Mountains.

Late Cretaceous ash flow volcanism and caldera formation in the Silver Bell Mountains [*Sawyer*, 1987] and in the Tucson Mountains [*Lipman*, 1989] produced locally thick sequences of volcanic and volcaniclastic rocks and regionally extensive ash flow sheets. The tuff of Confidence Peak was erupted from the Silver Bell caldera shortly before the Cat Mountain Tuff was erupted from the Tucson Mountains caldera. Paleomagnetic study of Upper Cretaceous rocks in the Silver Bell Mountains shows a 30° ± 16° clockwise rotation of this site relative to North America (*Hagstrum and Sawyer* [1989]; recalculated, see Table 3), and petrologic and age data [*Lipman*, 1991] indicate that the upper tuff unit capping the Silver Bell Mountains correlates with the Cat Mountain Tuff in the Tucson Mountains. This paleomagnetic study of the Tucson Mountains therefore was undertaken to determine a time-averaged paleomagnetic direction for the Tucson Mountains that could be compared with a reference direction for stable North America, and to evaluate the correlation of the Cat Mountain Tuff between the Silver Bell and Tucson Mountains.

Single cooling units of ash flow tuffs are usually magnetized over short periods of time relative to paleosecular variation (PSV) of the ancient geomagnetic field and have been correlated over large areas based on their uniform paleomagnetic directions [e.g., *Gromme et al.*, 1972; *Wells and Hillhouse*, 1989]. Thick intracaldera tuff units can have more variable paleomagnetic directions due to PSV and polarity reversals occurring during longer periods of cooling and magnetization [*Hagstrum and Lipman*, 1986; *R. L. Reynolds et al.*, 1986]. The eutaxitic structure of welded tuffs generally indicate the paleohorizontal [see *Hagstrum and Gans*, 1989], and between-site vertical axis rotations often can be determined from site mean directions after correction for structural tilt. In regions of crustal deformation characterized by rotations of fault blocks about near-vertical axes, however, correlations of tuff units can be supported only by site mean inclinations, because variations in declination are a direct measure of between-site vertical axis rotations. Thus the absolute rotations of the Silver Bell and Tucson Mountains, with respect to North America, can be tested by

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comparing them with the relative rotation between the two mountain ranges shown by the paleomagnetic declinations for a correlated Cat Mountain Tuff.

GEOLOGIC SETTING

Predominantly Upper Cretaceous volcanic rocks are exposed in the Tucson Mountains to the west of the city of Tucson in south central Arizona (Figure 2). These rocks are considered to be parts of the fill of a large ash flow caldera [*Lipman*, 1989] from which the rhyolitic Cat Mountain Tuff was erupted at about 72 Ma [*Bikerman and Damon*, 1966]; (recalculated ages from *S. J. Reynolds et al.*, [1986]). Previously called the Cat Mountain Rhyolite, the Cat Mountain Tuff was initially mapped as overlying a widespread breccia unit, the Tucson Mountain Chaos [*Brown*, 1939]. This unit has been the subject of much debate and has been variously interpreted as the sole of an imbricate thrust sheet [*Brown*, 1939], as a volcanically erupted intrusive breccia related to a regional "volcanic orogeny" [*Mayo*, 1963; 1971*a*], as stoped fragments rafted from deeper levels by rising pyroclastic magma which later erupted the Cat Mountain Tuff [*Bikerman*, 1963; *Mayo*, 1971*b*], and as part of a regional andesitic unit extending from at least the Silver Bell Mountains to the north to the Santa Rita and Sierrita Mountains to the south [*Drewes*, 1981].

Lipman [1976] compared the Tucson Mountain Chaos with caldera collapse breccias in the San Juan Mountains, southwestern Colorado, and inferred that the Tucson Mountain Chaos was also landslide breccia associated with collapse of a large ash flow caldera within which the Cat Mountain Tuff had ponded. The slide breccias in the Tucson Mountains have a matrix of nonwelded tuff, and rather than underlying the Cat Mountain Tuff, they interfinger with welded tuff at multiple horizons. These relations imply that the exposed thickness of the caldera-filling Cat Mountain Tuff is as much as 3 to 4 km [*Lipman*, 1989]. The breccias contain blocks up to 0.5 km across, vary in clast composition between caldera sectors, and consist of Paleozoic sedimentary rocks, Jurassic silicic volcanic rocks, and Cretaceous sedimentary and volcanic rocks [*Lipman*, 1989]. Included with the Cretaceous volcanic rocks are clasts of tuff of Confidence Peak which was erupted earlier from the Silver Bell caldera. Zones of decreased welding in the Cat Mountain Tuff are locally associated with horizons of slide breccias and also occur high in the unit, probably reflecting eruptive pauses or decreased thermal efficiency during emplacement of ash flows; reworked sediments or other depositional units that might indicate sizable time breaks have not been found along such zones of decreased welding. Thus the Cat Mountain Tuff appears to be a single thick intracaldera ash flow sheet with complex welding zonations. A small segment of the structural boundary of the Tucson Mountain caldera may be represented by an arcuate fault along the northwest flank of the mountain range which downdrops Cretaceous sedimentary rocks against Jurassic rocks (Figure 2).

Along the northeast and southeast flanks of the Tucson Mountains, the Cat Mountain Tuff is conformably overlain by a thick caldera-fill sequence of andesitic to rhyolitic lavas, with interbedded volcaniclastic sediments, deposited soon after caldera collapse [*Lipman*, 1989]. These rocks have shallow northeasterly dip angles (5°–25°) along the southeast flank of the range and steep northeasterly dip angles (50°–85°) along the northeast flank. The large zoned Amole pluton exposed in the northern Tucson Mountains (Figure 2) is a resurgent ring intrusion which cuts and arches the steeply dipping caldera fill upward to the east and north [*Lipman*, 1989]. K-Ar biotite ages of about 73 Ma for this intrusion [*Bikerman and Damon*, 1966] indicate that it was emplaced soon after eruption of the Cat Mountain Tuff, within analytical uncertainties of the isotopic ages. The volume of slide breccia is greater than the volume of ash flow tuff within the northern parts of the caldera fill, implying catastrophic subsidence in this sector. In contrast, megabreccia is virtually absent along the inferred southern caldera margin, and the thickness of the tuff decreases to about 100 m. These relations imply a hinged southern margin of the caldera and an overall trapdoor geometry of subsidence [*Lipman*, 1989]. A gridwork of northwest and northeast trending normal faults disrupts the caldera fill volcanic rocks in several parts of the mountain range; displacements along these faults are generally small (25–100 m), but are difficult to determine in the field due to the scarcity of stratigraphic markers. Some of these faults may be related to middle Tertiary extension, but others appear to have been associated with caldera subsidence in Late Cretaceous time [*Lipman*, 1989].

Overlying the caldera-related Cretaceous rocks of the Tucson Mountains are middle Tertiary volcanic rocks exposed on Tumamoc Hill and "A" Mountain at the southeastern side of the range, and in the vicinity of Safford Peak at the northern end of the range (Figure 2). The oldest unit of the Tumamoc Hill-"A" Mountain sequence is a distinctive plagioclase-phyric flow which has a K-Ar plagioclase age of about 29 Ma [*Bikerman and Damon*, 1966]. K-Ar whole rock and mineral ages for the overlying basaltic andesites and tuffs range from 28 Ma to 24 Ma for the uppermost lava flow of Tumamoc Hill [*Bikerman and Damon*, 1966; *Shafiqullah et al.*, 1980]. The sequence of dacitic flows and tuffs near Safford Peak, including the dacitic neck which forms Safford Peak itself, have similar K-Ar biotite ages spanning about 28 Ma to 25 Ma [*Bikerman and Damon*, 1966]. The middle Tertiary volcanic rocks generally dip gently to the northeast ($\simeq$10°–20°) and in the southeastern Tucson Mountains are basically conformable with the underlying Cretaceous rocks. This implies that most of the eastward tilting of the mountain range was associated with Tertiary faulting that may have been related to formation of the Santa Catalina core complex to the east [*Lipman*, 1989].

METHODS

Upper Cretaceous rocks in the Tucson Mountains were sampled for paleomagnetic analysis at 44 sites (Table 1). The sampling objectives were (1) to collect enough rock units, deposited over a sufficient span of time, to average PSV of the ancient geomagnetic field, (2) the Cat Mountain Tuff for correlation purposes, and (3) the Amole pluton to determine its attitude at the time of magnetization. Samples were collected with a portable gasoline-powered drill and were oriented in the field with both magnetic and solar compasses and a clinometer. Structural attitudes for the volcanic units are based on interbedded sedimentary rocks, local flow contacts, or eutaxitic structures for pyroclastic rocks. Data for the Amole pluton were corrected for the shallow northeasterly dip angles ($\simeq$13°) of nearby middle Tertiary lava

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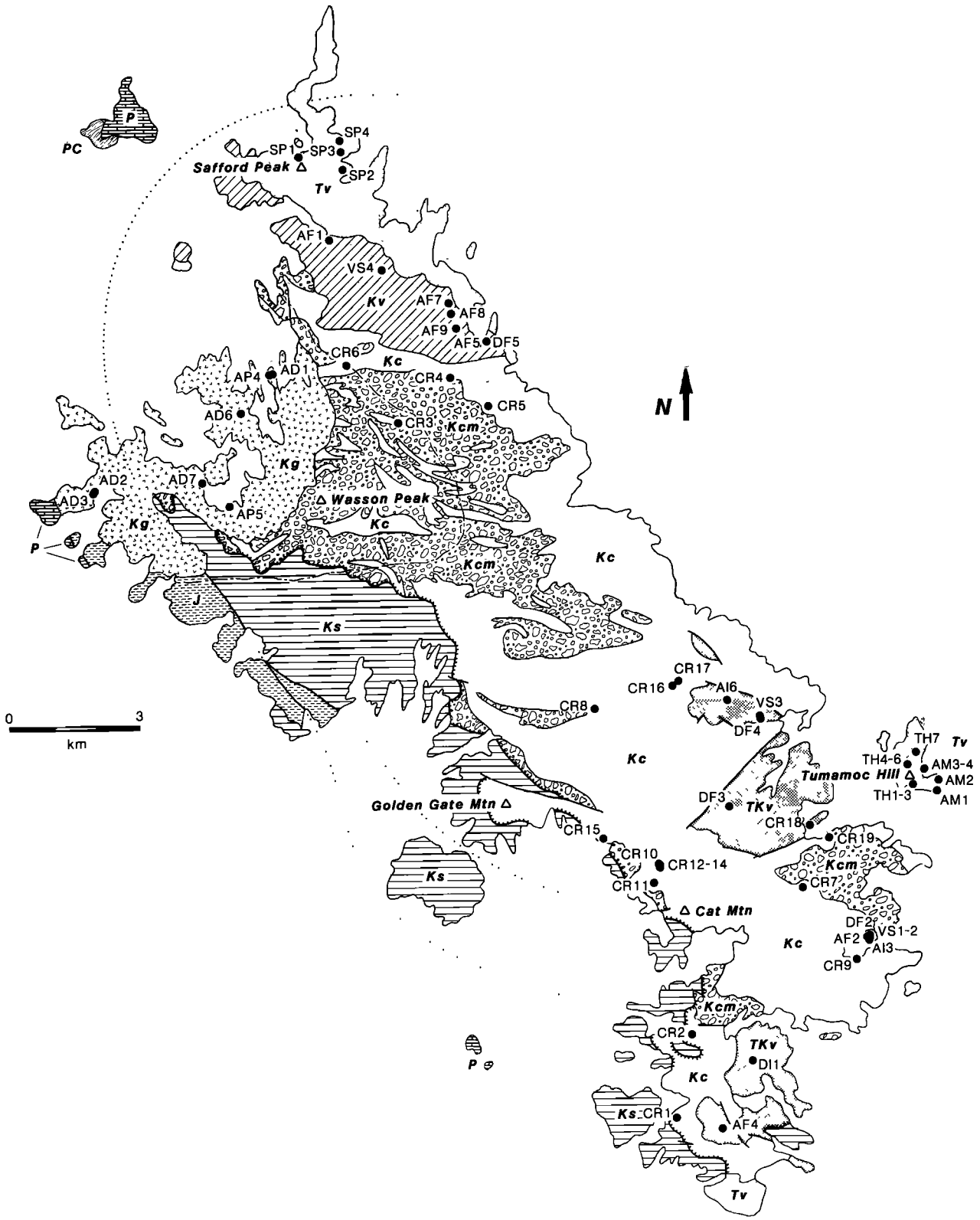


Fig. 2. Geologic map of the Tucson Mountains showing paleomagnetic sampling sites (see Tables 1 and 2). Map symbols are PC, Precambrian rocks; P, Paleozoic sedimentary rocks; J, Jurassic sedimentary and volcanic rocks; Ks, Cretaceous sedimentary rocks; Kc, Cretaceous Cat Mountain Tuff; Kcm, Cretaceous caldera collapse megabreccia; Kg, Cretaceous Amole pluton; Kv and TKv, Cretaceous and Tertiary caldera fill volcanic rocks; and Tv, Tertiary volcanic rocks. Fine lines indicate contacts; heavy lines indicate faults, dotted where concealed; hachured lines indicate floor of Tucson Mountains caldera; and broken line indicates outer limit of hornfels facies rocks surrounding the Amole pluton.

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TABLE 1. Site Mean Directions and Poles for Upper Cretaceous Units in the Tucson Mountains, Arizona

Site	Uncorrected		Strike/Dip	Corrected		N	R	k	α_{95}	Pole	
	I	D		I	D					Latitude	Longitude
					Amole Pluton						
AD1	77.7	302.6	315/13	74.2	355.1	5	4.8527	27	15.0	61.6	243.8
AD2	60.9	1.1	315/13	50.6	12.9	5	4.9693	130	6.7	79.0	340.3
AD3	65.9	8.8	315/13	54.6	20.4	6	5.8794	41	10.5	72.8	323.7
AP4	77.1	320.7	315/13	70.9	2.3	7	6.9713	209	4.2	67.0	252.2
AP5	82.5	350.1	315/13	71.7	25.2	8	7.9935	1082	1.7	60.2	277.0
AD6	65.5	302.0	315/13	65.2	330.7	8	7.8483	46	8.2	63.2	201.4
AD7	58.8	339.8	315/13	51.7	355.7	10	9.8535	61	6.2	86.4	160.9
Average				63.7	4.1	(7)	6.8468	39	9.8	76.6	261.3
					Dacitic Rocks						
DI1	−72.5	171.4	359/10	−71.1	202.0	8	7.9566	161	4.4	62.0	275.7
DF2	−57.6	216.3	267/25	−36.1	201.9	7	6.9551	134	5.2	67.0	5.4
DF3*	−12.5	141.3	300/20	−4.7	144.1	8	7.9797	345	3.0	45.0	124.8
DF4	−88.1	263.7	280/33	−56.4	193.3	9	8.9372	127	4.6	78.1	311.8
DF5	−81.6	339.4	300/25	−69.4	228.7	4	3.9919	371	4.8	49.7	293.2
					Volcaniclastic Sedimentary Rocks						
VS1	46.7	337.7	275/20	28.4	344.0	5	4.9041	42	12.0	67.6	113.2
VS2	51.0	328.4	267/25	28.1	337.0	5	4.8922	37	12.7	62.9	124.9
VS3	−78.8	90.8	280/33	−57.1	169.3	9	8.9543	175	3.9	79.9	194.0
VS4	58.2	202.8	278/72	48.4	356.3	7	6.7735	26	12.0	85.7	116.5
					Andesitic Rocks						
AF1	−54.4	320.6	260/60	−57.7	202.4	8	7.9695	229	3.7	70.8	314.2
AF2	62.6	47.9	267/25	43.0	26.2	4	3.9269	41	14.5	66.0	349.7
AI3	76.2	338.0	275/18	59.0	352.8	7	6.8488	40	9.7	80.4	213.8
AF4	61.5	348.6	359/10	61.7	7.2	7	6.9370	95	6.2	77.9	274.8
AF5	−79.1	7.6	300/25	−74.6	225.8	8	7.9652	201	3.9	48.8	280.7
AI6	−71.2	113.1	300/25	−61.0	168.7	5	4.9765	171	5.9	76.7	209.5
AF7	−76.0	79.2	282/60	−34.5	176.3	6	5.9660	147	5.6	76.2	83.8
AF8	−63.3	55.5	282/60	−46.8	165.1	8	7.9013	71	6.6	76.5	324.4
AF9	−54.6	67.1	282/60	−44.0	150.7	8	7.9497	139	4.7	63.6	152.2
					Cat Mountain Tuff						
CR1	−30.8	167.3	110/05	−35.0	165.5	8	7.9899	692	2.1	71.7	117.9
CR2	−31.4	160.7	360/05	−32.9	163.7	7	6.9917	721	2.3	69.5	118.8
CR3	−63.7	83.6	268/62	−26.8	148.4	5	4.9547	88	8.2	56.0	134.2
CR4	−31.5	78.5	283/75	−28.5	131.1	7	6.9706	204	4.2	42.5	149.7
CR5	−73.5	134.6	293/51	−31.5	185.0	8	7.9617	183	4.1	74.1	51.3
CR6†	−41.7	27.0	278/70	−63.0	155.6	5	4.9825	228	5.1	67.5	198.6
CR7†	−61.2	152.5	338/20	−57.1	186.2	8	7.9578	166	4.3	82.5	289.9
CR8	−52.5	146.2	288/30	−30.0	164.5	8	7.9893	655	2.2	68.6	113.6
CR9	−60.5	168.7	267/25	−35.6	172.0	8	7.9846	453	2.6	75.6	101.0
CR10†	−76.0	133.7	308/30	−55.9	192.6	8	7.9651	200	3.9	78.7	313.1
CR11	−54.5	146.3	308/30	−37.8	173.7	8	7.9837	429	2.7	77.7	97.7
CR12†	−71.3	149.2	308/30	−49.6	190.4	6	5.9933	750	2.5	80.9	347.2
CR13†	−71.9	127.0	308/30	−55.7	184.5	6	5.9959	1211	1.9	84.5	290.4
CR14†	−72.7	135.0	308/30	−54.0	187.8	6	5.9958	1194	1.9	83.1	317.3
CR15	−58.4	153.1	290/25	−38.4	170.8	8	7.9734	263	3.4	76.6	108.9
CR16†	−63.7	94.5	285/25	−57.9	140.0	6	5.9303	72	8.0	57.1	181.3
CR17†	−71.2	73.5	285/25	−68.3	147.1	6	5.9956	1146	2.0	59.3	207.5
CR18	−43.0	162.4	300/20	−28.2	172.2	8	7.9540	152	4.5	71.4	93.3
CR19	−51.4	170.4	300/20	−34.7	181.0	7	6.9736	228	4.0	76.9	64.7
Average of the lower unit				−34.2	170.4	(8)	7.9622	185	4.1	74.0	104.0
Average of the upper unit†				−54.5	188.4	(5)	4.9927	544	3.3	82.5	314.6
					All Units						
Average				55.6	1.2	(26)	24.7784	20	6.4	86.0	262.2

I and *D* are inclination and declination of paleomagnetic directions in degrees, uncorrected/corrected for structural tilt; Strike/Dip is strike and dip of unit (dip 90° clockwise of strike) at or near site in degrees; *N*, number of samples (sites) analyzed; *R* is vector sum of *N* unit vectors; *k* is precision parameter [*Fisher*, 1953]; α_{95} is radius of 95% confidence in degrees; Latitude and Longitude are latitude and east longitude of northern hemisphere virtual geomagnetic pole corrected for tilt in degrees; Average is mean of paleomagnetic directions (see text).

*Not included in overall mean.

†Upper magnetic unit of the Cat Mountain Tuff.

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TABLE 2. Site Mean Directions and Poles for Tertiary Units in the Tucson Mountains, Arizona

Site	Uncorrected		Strike/Dip	Corrected		N	R	k	α_{95}	Pole	
	I	D		I	D					Latitude	Longitude
					Tumamoc Hill						
TH1	59.9	294.5	301/15	58.3	319.7	5		[−112,−1]	[5.2,7.1]	56.9	182.2
TH2	57.6	253.2	301/15	66.7	276.4	7	6.9770	261	3.8	27.8	201.9
TH3	64.8	249.7	301/15	73.7	282.9	5	4.9656	116	7.1	33.8	212.7
TH4	70.8	246.7	301/15	78.9	297.0	8	7.9846	455	2.6	39.6	224.0
TH5	66.5	263.3	301/15	71.6	301.3	8	7.9702	235	3.6	43.5	208.3
TH6	73.9	287.3	301/15	70.9	335.5	6	5.9499	100	6.7	61.3	219.5
TH7	60.6	262.4	301/15	67.0	290.6	6	5.8780	41	10.6	36.9	199.8
Average				70.7	301.5	7	6.9135	69	7.3	43.7	206.4
					"A" Mountain						
AM1	64.7	321.8	301/15	56.5	344.6	6	5.8405	31	12.2	76.5	183.9
AM2	−47.8	162.1	301/15	−37.0	171.7	8	7.9865	517	2.4	76.2	103.5
AM3	−48.8	155.6	301/15	−39.1	166.7	7	6.9110	67	7.4	74.4	121.5
AM4	−44.5	182.8	301/15	−31.0	187.9	7	6.9573	140	5.1	72.9	42.5
Average				41.2	353.7	4	3.9158	36	15.6	79.7	103.3
					Safford Peak						
SP1*	23.6	173.1	315/13	31.2	167.6	7	6.9716	211	4.2	39.4	84.3
SP2	−41.1	176.1	315/13	−32.0	183.0	9	8.9077	87	5.6	74.7	57.9
SP3	−45.3	170.3	315/13	−36.9	179.0	7	6.9723	216	4.1	78.2	73.4
SP4	−60.0	167.4	315/13	−51.5	182.1	7	6.9458	111	5.8	88.2	342.8
Average				40.1	1.3	3	2.9679	62	15.8	80.5	61.3

See Table 1 notes. Bingham statistics [*Onstott*, 1980] for site TH1: R is not calculated, $[-k_1, -k_2]$ are the two Bingham concentration parameters, and $[\alpha_1, \alpha_2]$ are the two Bingham 95% confidence limits.
*Not included in overall mean.

flows, and available local or regional corrections were used for the other intrusive units. Structural corrections were made by a simple rotation of the inferred paleohorizontal to the present horizontal about the strike axis.

The Cat Mountain Tuff was initially sampled at sites in the southern Tucson Mountains to avoid potential structural complications associated with the more intensely deformed rocks to the north and to avoid thick sections that might have cooled more slowly and recorded significant PSV. Site CR6 was collected near the eastern margin of the Amole pluton (Figure 2) to see if these rocks had been remagnetized by the resurgent intrusion. Preliminary results implied the presence of two distinct magnetic units within the Cat Mountain Tuff, so further sampling was undertaken in nonwelded to densely welded portions of the tuff to refine the paleomagnetic directions and local stratigraphy of these two units. Postcaldera andesitic and dacitic lava flows were sampled at 11 geographically dispersed sites in order to sample the largest possible number of individual units representing the broadest possible range of time. In addition, intrusive bodies of similar composition were sampled at sites AI3, AI6, and DI1, and intercalated volcaniclastic units were sampled at sites VS1-4 (Table 1). The finer-grained granodioritic border phase (AP4-5) and late-stage aplite dikes (AD1-3 and AD6-7) of the Amole pluton were sampled at seven sites. The middle Tertiary volcanic rocks were samples at 11 sites in the vicinity of Tumamoc Hill and at five sites in the vicinity of Safford Peak (Table 2).

The remanent magnetization of individual specimens cut from the core samples was measured with a cryogenic magnetometer. Specimens from all samples were subject to progressive alternating field (af) demagnetization, and lines representing the characteristic magnetization directions were fitted to the demagnetization data using a least squares method based on principal component analysis [*Kirschvink*, 1980]. The statistical methods of *Fisher* [1953] were used in analyzing most within-site and unit means. Bingham statistics [*Onstott*, 1980] were calculated for the site mean direction (TH1, Table 2) determined using remagnetization circle analysis [*Halls*, 1978]. Estimates of inclination flattening and clockwise rotation of the paleomagnetic directions for sites in southern Arizona, with respect to North American reference or correlated tuff unit directions, and their 95% confidence limits (Table 3) were calculated after the methods of *Beck* [1980], as modified by *Demarest* [1983].

RESULTS

Natural remanent magnetization (NRM) directions and intensities for all samples from the Tucson Mountains were measured prior to af demagnetization treatments. Within-site NRM directions for samples of the Cat Mountain Tuff were typically well grouped but were moderately to strongly dispersed for samples from sites CR2–CR5 and CR19. The dispersed directions, in association with intensities 1–3 orders of magnitude higher than those for other samples of the tuff, indicate that samples from these five sites may have been affected by surface currents caused by direct or nearby lightning strikes. Excepting these five sites, the NRM intensities for sites in the Cat Mountain Tuff form two distinct populations: one with an estimated average intensity of 2.09×10^{-2} A m^{-1} and corresponding standard deviation (s) of 9.76×10^{-3} A m^{-1}, and another with an estimated average intensity of 2.28×10^{-1} A m^{-1} ($s = 1.75 \times 10^{-1}$ A m^{-1}). Af demagnetization was successful in removing presumed lightning-induced magnetizations from samples of the Cat Mountain Tuff and in isolating their reversed-polarity characteris-

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TABLE 3. Vertical Axis Rotations of Upper Cretaceous and Tertiary Rocks in South Central Arizona

	I	D	N	R	k	α_{95}	d	r	Reference
				Tucson Mountains					
Upper Cretaceous rocks									
North America	57.5	353.8	(26)	24.5310	18	7.1			1
All units	55.6	1.2	(26)	24.7784	20	6.4	1.9 ± 7.5	7.4 ± 13.7	4
North America	57.5	353.8	(26)	24.5310	18	7.1			1
Amole pluton	63.7	4.1	(7)	6.8468	39	9.8	−6.2 ± 9.4	10.3 ± 20.4	4
Middle Tertiary rocks									
North America	49.9	352.2	(102)	94.6429	14	3.9			3
All units	40.8	357.0	(7)	6.8747	48	8.8	9.1 ± 7.5	4.8 ± 10.2	4
				Silver Bell Mountains					
Upper Cretaceous rocks									
North America	57.6	353.8	(26)	24.5310	18	7.1			1
All units	61.2	24.0	(34)	31.1472	12	7.6	−3.6 ± 8.1	30.2 ± 16.2	2
Middle Tertiary rocks									
North America	50.2	352.2	(102)	94.6429	14	3.9			3
All units	43.8	357.3	(10)	9.0541	10	16.5	6.4 ± 13.3	5.1 ± 18.5	2
				Cat Mountain Tuff					
Tucson Mountains	−34.2	170.4	(8)	7.9622	185	4.1			4
Silver Bell Mountains	−38.7	187.3	(4)	3.9705	102	9.2	−4.5 ± 7.8	17.0 ± 10.0	2

Abbreviations as in Table 1. Parameters d and r are inclination flattening and clockwise rotation at site with respect to expected field direction, in degrees, with 95% confidence limits after *Beck* [1980], as modified by *Demarest* [1983]. References are 1, expected direction at site calculated from pole at 82°N, 210°E, *Gunderson and Sheriff* [1991]; 2, *Hagstrum and Sawyer* [1989]; 3, calculated from pole at 83°N, 148°E, *Diehl et al.* [1983]; and 4, this study.

tic magnetizations (Figure 3*a*). Site mean directions for the tuff are listed in Table 1 and are shown both before and after application of structural corrections in Figure 4. The dispersion of uncorrected directions in Figure 4*a* generally reflect the increasing dip angles of the Upper Cretaceous units to the north in the Tucson Mountains, and the corrected directions (Figure 4*b*) show the presence of two populations, indicating that the Cat Mountain Tuff can be subdivided into two distinct magnetic units. Furthermore, the two populations of sites with different directions coincide with those of

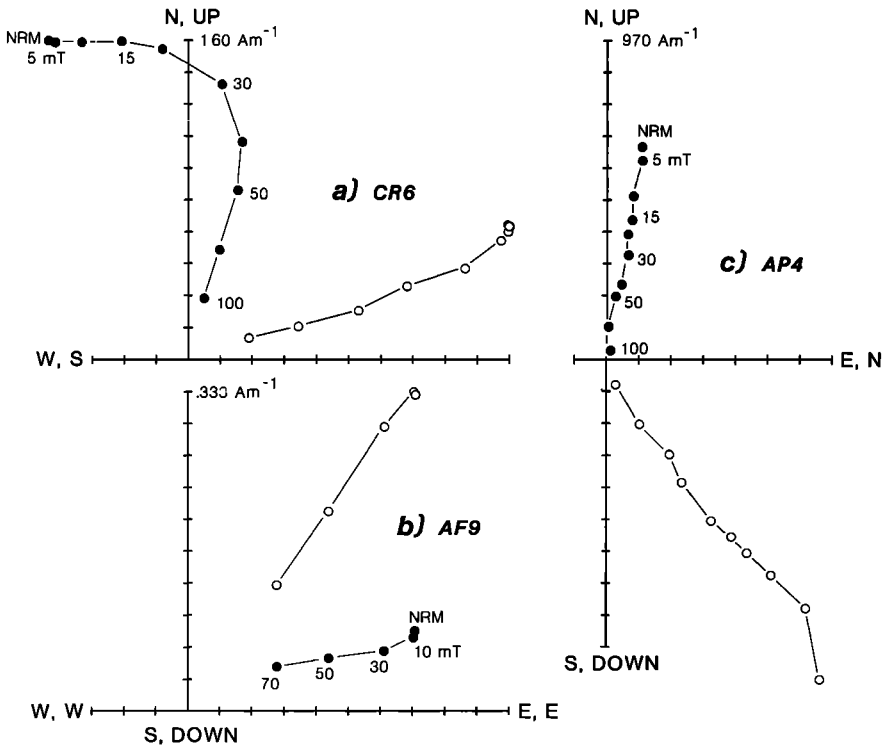


Fig. 3. Orthogonal projections of uncorrected alternating field demagnetization vector endpoints for samples of (*a*) Cat Mountain Tuff from site CR6, (*b*) caldera fill andesitic lava flow from site AF9, and (*c*) granodioritic border phase of the Amole pluton from site AP4. Solid symbols indicate projections of vector endpoints onto the horizontal plane and open symbols onto the vertical plane.

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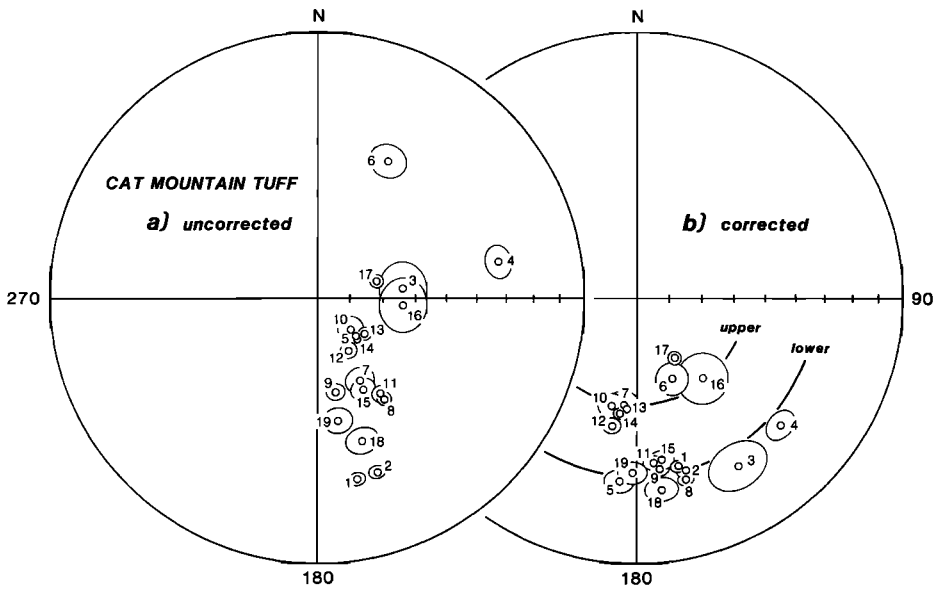


Fig. 4. Mean directions of sites with 95% confidence limits (*a*) uncorrected and (*b*) corrected for structural tilt for the Cat Mountain Tuff (Table 1). Small-circle arcs represent the average inclination values for the lower and upper magnetic units. Diagram is an equal-area plot in which open symbols indicate projection from the upper hemisphere.

NRM intensity, implying slight differences in quantity or size of the magnetic minerals between the two units of the tuff. The dispersion of the site mean inclinations [*McFadden and Reid*, 1982] decreases significantly (99% confidence level [*McElhinny*, 1964]) after correction for each unit, demonstrating that the remanent magnetizations predate tilting of the Cat Mountain Tuff. Because dispersion of the corrected site mean declinations along small-circle arcs (Figure 4*b*) most likely depends on the magnitude of inclined-axis rotation during deformation and therefore on the present dip angle [see *Chan*, 1988], sites with dip angles >45° are deemed unreliable. At less deformed sites this dispersion also could have resulted from local vertical axis rotations (e.g., CR16 and CR17). Samples from sites CR3–CR5 (lower unit) and sites CR6, CR16, and CR17 (upper unit) thus were omitted from unit mean calculations for the two magnetic units (Table 1).

The andesitic, dacitic, and volcaniclastic rocks of the caldera fill sequence also showed the probable effects of lightning-induced magnetizations, but af demagnetization was successful in determining the characteristic directions of magnetization for these rock units as well (e.g., Figure 3*b*). The caldera fill rocks have both normal- and reversed-polarity magnetizations (Figure 5), indicating that they were deposited over a significant period of time relative to PSV. Site DF3 is in a massive structureless flow(?), and a regional attitude was used for the structural correction. The inconsistency of the corrected direction for site DF3 with the other corrected directions (Table 1), however, implies that either the regional attitude is locally incorrect or the unit was magnetized in an anomalous geomagnetic field (i.e., excursion or reversal); this site therefore was omitted from the overall mean calculation.

Upon af demagnetization (Figure 3*c*), samples from the Amole pluton show consistent normal-polarity characteristic directions of magnetization (Table 1). The angular standard deviation value for the unit mean pole of the Amole pluton (S = 17.6°) is similar to the expected value (S = 15.5° ± 1°) for the latitude of the Tucson Mountains, based on models of recent PSV [*McFadden et al.*, 1988], and we infer that the pluton cooled and was magnetized over a sufficient period of time to average PSV. Normal- and reversed-polarity characteristic magnetizations for the middle Tertiary units, although limited in number, also indicate that these rocks were emplaced over a significant time interval with respect to PSV (Figure 5). Lightning-induced magnetizations also appear prevalent in these rocks, and one site in a lava flow near Safford Peak was completely overprinted and therefore discarded; the mean direction for the uppermost site on Tumamoc Hill (TH1) could only be determined using remagnetization circle analysis (Table 2 and Figure 6).

Dispersion of the site mean directions for Upper Cretaceous units in the Tucson Mountains (not including those for the Cat Mountain Tuff) is significantly decreased at the 95% confidence level [*McElhinny*, 1964] after structural correction, indicating that the remanent magnetizations predate deformation represented by these corrections (Table 1 and Figure 5). The angular standard deviation for the overall Late Cretaceous paleomagnetic pole for the Tucson Mountains (S = 20.8°) is larger than the expected model value based on PSV alone. The excessive dispersion is probably due to errors associated with the structural corrections; these errors are assumed to be nonsystematic and not to have significantly affected the overall mean paleomagnetic direction. The averaging of PSV by the paleomagnetic data for the Tucson Mountains is also supported by antipodal means for the normal-polarity (I = +55.7°, D = 359.2°, α_{95} = 9.5°, N = 13) and reversed-polarity (I = −55.6°, D = 182.9°, α_{95} = 9.7°, N = 13) site mean directions (Figure 5).

Discussion

Tucson Mountains

Mauger et al. [1965] suggested a regional correlation of the Cat Mountain Tuff in the Tucson Mountains with the tuff unit

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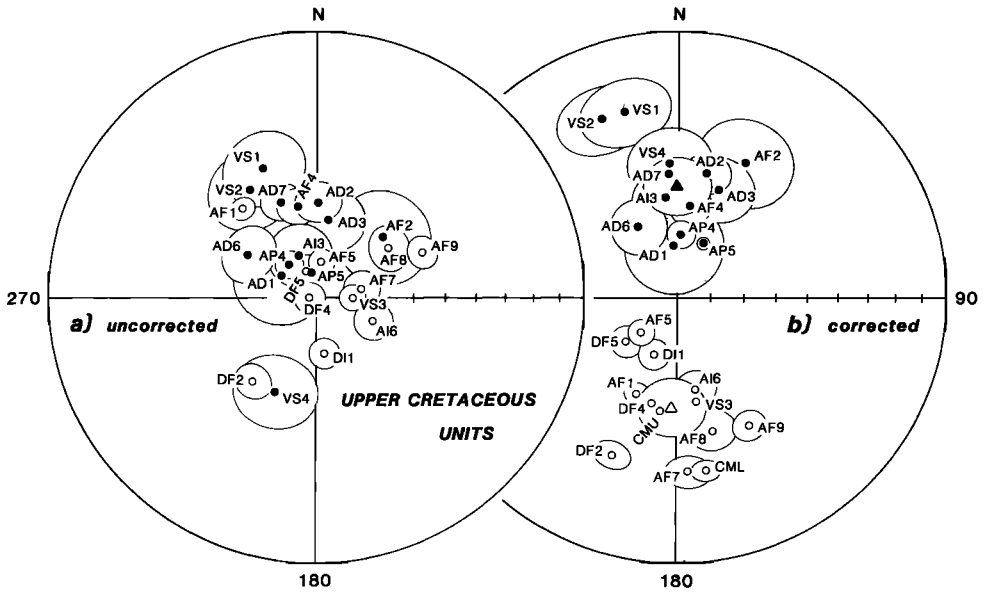


Fig. 5. Mean directions of sites with 95% confidence limits (*a*) uncorrected and (*b*) corrected for structural tilt for Upper Cretaceous units in the Tucson Mountains (Table 1). CMU and CML indicate means for the upper and lower magnetic units of the Cat Mountain Tuff, respectively. Triangular symbols indicate mean directions of the normal (N = 13) and reversed polarity (N = 13) site mean directions. Diagrams are equal-area plots in which solid symbols indicate projection from the lower hemisphere and open symbols from the upper hemisphere.

of the Mount Lord Volcanics, which caps the Silver Bell Mountains to the northwest (Figure 1). Although *Mauger et al.* [1965] reported a K-Ar sanidine age of 60.9 ± 1.8 Ma for the tuff unit of the Mount Lord Volcanics, which is significantly younger than that for the Cat Mountain Tuff in the Tucson Mountains (71.9 ± 2.1 Ma [*Bikerman and Damon*, 1966]), thin section observations showed that the apparent age of the sample from the Silver Bell Mountains most likely was reduced by argon diffusion related to a younger episode of alteration and mineralization [*Mauger et al.*, 1965]. Recently acquired petrologic data also support the correlation of the Cat Mountain Tuff between the Silver Bell and Tucson Mountains; in particular, the Cat Mountain Tuff has distinctively high zirconium contents (100–400 ppm [*Lipman*, 1991]). In addition, the paleomagnetic data for the Cat Mountain Tuff presented here and by *Hagstrum and Sawyer* [1989] support this correlation. A mean inclination calculated from the data for samples least affected by deformation or probable lightning strikes (sites CM1–CM4) for the tuff unit in the Mount Lord Volcanics of the Silver Bell Moun-

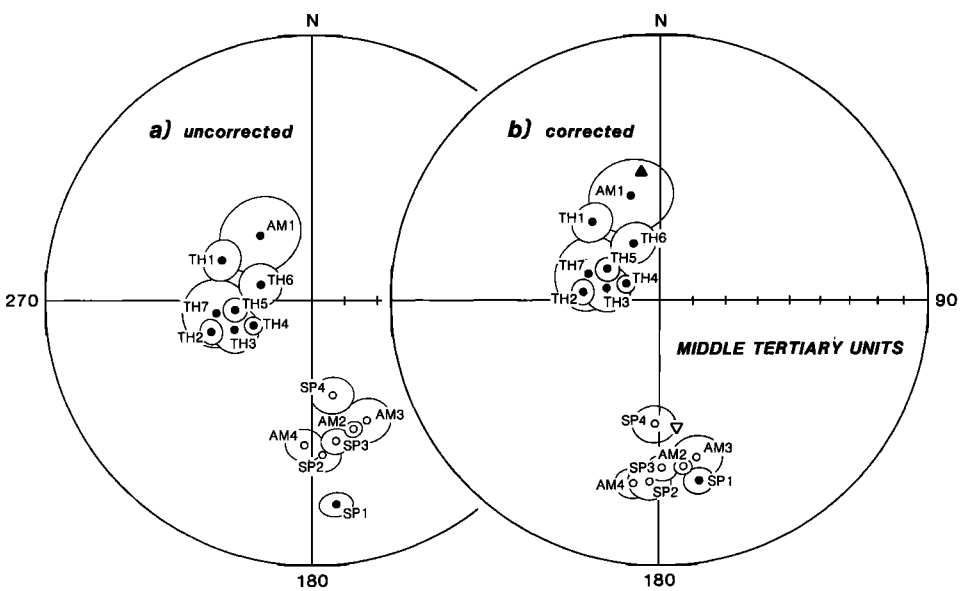


Fig. 6. Mean directions of sites with 95% confidence limits (*a*) uncorrected and (*b*) corrected for structural tilt for middle Tertiary units (Table 2). Triangular symbols indicate expected middle Tertiary directions for the Tucson Mountains. Diagram is an equal area plot in which solid symbols indicate projection from the lower hemisphere and open symbols from the upper hemisphere.

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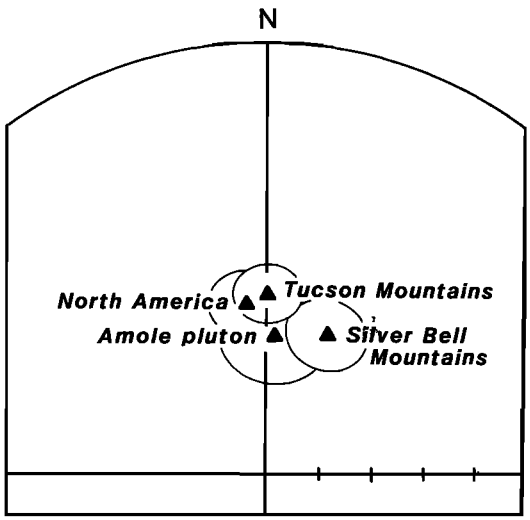


Fig. 7. Mean directions with 95% confidence limits for Upper Cretaceous units in the Tucson and Silver Bell Mountains and for the Amole pluton. Late Cretaceous reference direction for North America calculated from *Gunderson and Sheriff* [1991] (see Table 3). Diagram is an equal-area plot in which solid symbols indicate projection from the lower hemisphere.

tains (I = −38.7° ± 9.2°) is similar to that of the lower magnetic unit of the Cat Mountain Tuff (I = 34.2° ± 4.1°) in the Tucson Mountains (Table 3), and these inclinations are quite different than the average expected inclination for the region in Upper Cretaceous time (±58°). Furthermore, the mean declinations indicate a 17° ± 10° clockwise rotation of the Silver Bell Mountains relative to the Tucson Mountains (Table 3 and Figure 7). The 13° ± 19° difference between this 17° relative rotation between mountain ranges and the absolute clockwise rotation of the Silver Bell Mountains relative to North America (30°), although insignificant at the 95% confidence level, implies some clockwise rotation of the Tucson Mountains as well.

Comparison of the overall mean direction for Upper Cretaceous units in the Tucson Mountains (I = +55.6°, D = 1.2°, α_{95} = 6.4°, N = 26) to a reference direction calculated for contemporaneous units from stable North America (I = +57.5°, D = 353.8°, α_{95} = 7.1°, N = 26) indicates a slight northward shift (2° ± 8°) and clockwise rotation (7° ± 14°) of the site since Late Cretaceous time, although both these values are insignificant with 95% confidence (Table 3 and Figure 7). The magnitudes of the absolute rotations for the Tucson and Silver Bell Mountains strongly depend on the position of the North American reference pole, and because of rapid apparent wander in the Late Cretaceous, an accurate pole for this time has been difficult to determine. The best available estimate of a Late Cretaceous reference pole for North America is that recently published by *Gunderson and Sheriff* [1991] based on volcanic rocks in Montana (Table 3). No significant latitudinal shift relative to stable North America is expected for the Tucson Mountains since this time, and we attribute the observed inclination flattening value to error. The 7° ± 14° clockwise rotation of the Tucson Mountains, however, is of the same sense as the rotation (30° ± 16°) reported for the Silver Bell Mountains [*Hagstrum and Sawyer*, 1989] and is consistent with the 17° ± 10° relative rotation between mountain ranges indicated by paleomagnetic directions for the Cat Mountain Tuff. Thus we infer that the Tucson Mountains have probably undergone a clockwise vertical axis rotation of about 10°.

Similarity of the mean direction for the 73-Ma-old Amole pluton with the overall mean direction for the Tucson Mountains (Table 3), after correction for the tilt of the overlying volcanic units (≈13° NE, Table 2), indicates that most of the tilting of the northern Tucson Mountain block since Late Cretaceous time occurred after deposition of the middle Tertiary volcanic rocks. Steep tilting to the north and northeast of the caldera fill adjacent to the Amole pluton, however, occurred before the pluton cooled and acquired its characteristic magnetization in Late Cretaceous time. Thus these data support the interpretation that the caldera fill sequence was deformed by emplacement of the Amole pluton and that the pluton represents the solidified remnants of the magma chamber which underlay the Tucson Mountains caldera at the time of collapse and resurgent uplift. Although site CR6 in the Cat Mountain Tuff is within ≈600 m of the contact with the Amole pluton (Figure 2), samples from this site have not been significantly overprinted by a thermoremanent magnetization related to the nearby emplacement of the pluton; hornfels alteration of the country rock appears to be concentrated along the southeastern margin of the pluton in the vicinity of Wasson Peak.

The mean direction for site CR6 is among the group of mean directions with steeper inclinations (Table 1 and Figure 4*b*), indicating that this site is within the upper magnetic unit of the Cat Mountain Tuff. The identification of two distinct magnetic units within the intracaldera Cat Mountain Tuff is unusual because thick caldera-fill sequences of tuff usually cool over longer periods of time and record greater amounts of PSV [e.g., *Hagstrum and Lipman*, 1986; *R. L. Reynolds et al.*, 1986]. Rapid cooling of the thick (3 km) intracaldera Cat Mountain Tuff, composed predominantly of the lower magnetic unit, however, probably resulted from the introduction of large quantities of cold landslide-related megabreccia during its emplacement. This rapid cooling is corraborated by the nonwelded matrix of the intracaldera megabreccia. The ≈20° difference in inclination between the mean directions for the two magnetic units, however, may indicate that a significant amount of time (10^2–10^3 years) passed between magnetization of these units based on estimates of recent rates of secular variation [e.g., *Merrill and McElhinny*, 1983].

The two magnetic units in the upper part of the Cat Mountain Tuff could alternatively represent (1) the presence of two discrete depositional units for which the contact is concealed or (2) rapid PSV during laterally variable cooling of a single tuff unit. The first interpretation is favored by the bimodal distribution of inclinations and NRM intensities and by the observation of high and low inclinations in nonwelded to densely welded tuff. If this interpretation is correct, the paleomagnetic data for the intracaldera Cat Mountain Tuff provide criteria for inferring a concealed stratigraphic break and the presence of fault displacements sufficient to juxtapose the two units in a complicated pattern throughout the Tucson Mountains. The second interpretation of variable cooling within the upper part of a single depositional unit is favored by the observation of high (CR12–CR14, CR16, and CR17) and low (CR9 and CR18) inclinations in the upper nonwelded to partly welded zone of the tuff, where mapped

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as immediately overlain by sedimentary rocks. Because no zoned of nonwelded to partly welded tuff is exposed at lower stratigraphic levels, it is difficult to account for the juxtaposition of two similar-appearing units at the paleomagnetic sampling sites by sizeable faults. Resolution of these alternatives, however, is not critical to the major interpretations of this study concerning regional vertical axis rotations.

Although the middle Tertiary volcanic units in the Tucson Mountains were deposited and magnetized over a time span including both normal- and reversed-polarity fields (Table 2 and Figure 6), the eruptive rates over this time span apparently were irregular. All lava flows sampled near Safford Peak show reversed-polarity magnetizations and were likely deposited within a single polarity chron, even though ash flow tuffs and volcanic sediments between dacite flows indicate some eruptive pauses between the lavas. The volcanic sequence, exposed on Tumamoc Hill and "A" Mountain, consists of lower and upper packets of basaltic andesite lavas separated by sediments and a weakly welded rhyolitic ash flow sheet. A polarity reversal is recorded within the lower packet, between eruption of the basal "turkey track" porphyritic andesite (AM1) and overlying flows of aphyric andesite (AM2 and AM3). The ash flow tuff (AM4) is similarly of reversed polarity, but seven flows of the upper lava packet (TM1–TM7) all have distinctive normal-polarity magnetizations (Figure 6*b*) that likely indicate rapid eruption during an interval of high-amplitude PSV; a $\simeq$70° counterclockwise rotation of these rocks is inconsistent with data for either "A" Mountain or Upper Cretaceous units in the southern Tucson Mountains. The mean direction for the dacite neck forming Safford Peak also probably reflects extreme PSV and was omitted, along with site means for Tumamoc Hill, from calculation of the overall middle Tertiary mean direction for the Tucson Mountains (Table 3). Although the remaining data (N = 7) are too few to completely average PSV, the general agreement of these data with those for the Silver Bell Mountains and with the expected North American direction (Table 3) is consistent with no significant vertical axis rotation of the Tucson Mountains after middle Tertiary time.

Regional Deformation

The clockwise vertical axis rotations inferred for the Silver Bell Mountains [*Hagstrum and Sawyer*, 1989], Tucson Mountains, and Canelo Hills region [*Hagstrum and Lipman*, 1991] imply that regional rotational deformation has occurred in southern Arizona since Late Cretaceous time. Although insufficient in number to convincingly average PSV, the paleomagnetic data for middle Tertiary units in the Silver Bell and Tucson Mountains imply that clockwise rotation of these mountain ranges preceded deposition of the Tertiary units. The clockwise rotations of the Silver Bell and Tucson Mountains, thus interpreted to be of Late Cretaceous to early Tertiary age, could be part of a larger pattern of clockwise rotations which have occurred since middle Cretaceous time along the western margin of the North American Cordillera [*Beck*, 1988]. Although no one mechanism can explain all of these rotations [*Cox*, 1980], most are clockwise [*Beck*, 1980] and therefore are possibly linked to regional dextral shear along the western margin of ancestral North America caused by northward oblique subduction of the Kula and Farallon plates [*Cox*, 1980; *Beck*, 1983; *Engebretson et al.*, 1984]. Paleomagnetic data for the Baja California Peninsula to the southwest [e.g., *Hagstrum and Filmer*, 1990] and for the Colorado Plateau to the north [*Steiner*, 1988; *Bryan and Gordon*, 1990; see also *Hamilton*, 1988] also indicate that clockwise rotations have occurred in southwestern North America since middle Cretaceous time, supporting the evidence for similar deformation in southern Arizona during Late Cretaceous and early Tertiary time. The shallow angle of subduction [e.g., *Dickinson and Snyder*, 1978; *Bird*, 1988] and rapid convergence rate [*Engebretson et al.*, 1984] inferred for the Farallon plate during this time would also contribute to the likelihood of rotational deformation in southern Arizona.

Mechanisms of vertical axis rotation considered by *Nelson and Jones* [1987] for the Las Vegas Valley shear zone also potentially apply to the rotations observed in southern Arizona and include (1) pervasive, continuous simple shear, (2) block rotation between conjugate shear faults, and (3) rotation of individual blocks in response to distributed shear in the ductile lower crust and mantle. Pervasive simple shear can be quickly eliminated for southern Arizona because the Upper Cretaceous rocks there do not show any evidence of extensive internal deformation or recrystallization that would be associated with this mechanism. Furthermore, the available paleomagnetic data for southern Arizona imply that the Silver Bell and Tucson Mountains each rotated as a rigid block. In models of rotation of conjugate shear faults, domains of rigid crustal blocks bounded by parallel strike-slip faults rotate along with the faults [e.g., *Ron et al.*, 1984; *McKenzie and Jackson*, 1986]. Paleomagnetic results for the lower unit of the Cat Mountain Tuff show that the individual rotations of the Silver Bell and Tucson Mountains are significantly different (17° $\pm$ 10°), implying that either the conjugate shear fault model is incorrect for southern Arizona or that the two mountain ranges were in different rotational domains [e.g., *Garfunkel and Ron*, 1985]. Vertical axis rotations of individual crustal blocks, driven by ductile shear below, depend upon the shape and orientation of the individual blocks [*Lamb*, 1987] and could explain the variable amount and sense of rotation observed in south central Arizona. Much more data is needed, however, before a well-constrained model of vertical axis rotations associated with Laramide deformation in south central Arizona can be constructed.

CONCLUSIONS

Comparisons of paleomagnetic data for Upper Cretaceous rocks in the Silver Bell [*Hagstrum and Sawyer*, 1989], Tucson Mountains, and Canelo Hills region [*Hagstrum and Lipman*, 1991] imply that all of these localities in south central Arizona have been rotated about vertical axes with respect to cratonic North America. The inference that rocks within the Basin and Range province of southern Arizona have all remained unrotated with respect to stable North America since Jurassic time thus is unfounded. Apparently concordant middle Tertiary paleomagnetic data for the Silver Bell Mountains [*Hagstrum and Sawyer*, 1989] and the Tucson Mountains (this study) imply that these clockwise rotations, at least, occurred during latest Cretaceous to early Tertiary time. *Hagstrum and Sawyer* [1989] suggested that the clockwise rotation of the Silver Bell Mountains was related to right lateral strike-slip displacement on northwest

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trending faults bounding the Silver Bell block. Strike-slip faults most likely played an important role in the Laramide deformation of southern Arizona [*Keith and Wilt*, 1986], although evidence pertaining to their role has been difficult to obtain because of poor exposures in the region and the lack of local features indicating the sense and amount of displacement. The Silver Bell and Tucson blocks have been rotated clockwise different amounts, apparently in response to continuous ductile deformation in the lower crust and mantle associated with oblique subduction along the continental margin, although much more data is needed to determine the exact mechanism(s) of deformation. This rotational deformation is coincident with rapid plate convergence and Laramide compressional stresses and also could be related in part to inhomogeneous shortening of the crust in southern Arizona [*Davis*, 1979, 1981].

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